

# JIE XUE

## PERSONAL DATA

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## EMPLOYMENT

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| AUG 2021 - CURRENT   | <i>Assistant Professor of Computer Science</i><br><b>New York University Shanghai, China</b>                                                |
| SEPT 2019 - AUG 2021 | <i>Postdoctoral Scholar</i><br><b>University of California, Santa Barbara, USA</b><br>Hosts: Prof. Subhash Suri and Prof. Daniel Lokshantov |
| JAN 2015 - MAY 2018  | <i>Teaching Assistant (part time)</i><br><b>University of Minnesota, Twin Cities, USA</b>                                                   |

## EDUCATION

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| SEPT 2014 - JULY 2019 | <i>Doctoral Degree</i><br><b>University of Minnesota, Twin Cities, USA</b><br>Major: Computer Science<br>Minor: Mathematics<br>Research Interests: Computational Geometry<br>Advisor: Prof. Ravi Janardan<br>GPA: 4.0/4.0         |
| SEPT 2011 - APR 2014  | <i>Master's Degree</i><br><b>Nanjing University of Aeronautics &amp; Astronautics, China</b><br>Major: Computer Technology<br>Research Interests: Data Mining, Pattern Recognition<br>Advisor: Prof. Songcan Chen<br>GPA: 2.2/4.0 |
| SEPT 2008 - JUNE 2011 | <i>Bachelor's Degree</i><br><b>Nanjing University of Aeronautics &amp; Astronautics, China</b><br>Major: Computer Science & Technology<br>GPA: 1.9/4.0                                                                            |

## RESEARCH INTERESTS

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- Theoretical Computer Science
  - Computational Geometry
  - Algorithms & Data Structures
  - Graph Theory
  - Parameterized Complexity

## PUBLICATIONS

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(Authors with \* are sorted by  $\alpha$ - $\beta$  order. Others are sorted by contribution.)

### Surveys and book chapters

1. **Jie Xue\***, Meirav Zehavi\*, Parameterized algorithms on geometric intersection graphs. To appear in *Computer Science Review*, 2025.

### Journal publications

1. Timothy M. Chan\*, Qizheng He\*, Subhash Suri\*, **Jie Xue\***, Dynamic geometric set cover, revisited. In *SIAM Journal on Computing*, 2025.
2. Kyungjin Cho\*, Eunjin Oh\*, Haitao Wang\*, **Jie Xue\***, Optimal algorithm for the planar two-center problem. In *TheoretCS* (invited from SoCG'24), 2024.
3. Haitao Wang\*, **Jie Xue\***, Algorithms for halfplane coverage and related problems. In *Discrete & Computational Geometry*, 2024.
4. Sayan Bandyapadhyay\*, William Lochet\*, Daniel Lokshantov\*, Saket Saurabh\*, **Jie Xue\***, True contraction decomposition and almost ETH-tight bipartization for unit-disk graphs. In *ACM Transactions on Algorithms*, 2024.
5. Akanksha Agrawal\*, Tanmay Inamdar\*, Saket Saurabh\*, **Jie Xue\***, Clustering what matters: optimal approximation for clustering with outliers. In *Journal of Artificial Intelligence Research*, 2023.
6. Pankaj K. Agarwal\*, Hsien-Chih Chang\*, Subhash Suri\*, Allen Xiao\*, **Jie Xue\***, Dynamic geometric set cover and hitting set. In *ACM Transactions on Algorithms*, 2022.
7. **Jie Xue**, Yuan Li, Rahul Saladi, Ravi Janardan, New bounds for range closest-pair problems. In *Discrete & Computational Geometry*, 2022.
8. Haitao Wang\*, **Jie Xue\***, Improved algorithms for the bichromatic 2-center problem for pairs of points. In *Computational Geometry: Theory and Applications*, 2021.
9. **Jie Xue**, Yuan Li, Rahul Saladi, Ravi Janardan, Searching for the closest-pair in a query translate. In *Journal of Computational Geometry* (SoCG'19 special issue), 2020.
10. Haitao Wang\*, **Jie Xue\***, Near-optimal algorithms for shortest paths in weighted unit disk graphs. In *Discrete & Computational Geometry* (SoCG'19 special issue), 2020.
11. Timothy Chan\*, Rahul Saladi\*, **Jie Xue\***, Range closest-pair search in higher dimensions. In *Computational Geometry: Theory and Applications* (WADS'19 special issue), 2020.
12. **Jie Xue**, Yuan Li, Ravi Janardan, Approximate range closest-pair queries. In *Computational Geometry: Theory and Applications* (CCCG'18 special issue), 2020.
13. Yuan Li, Ahmed Eldawy, **Jie Xue**, Nadezda Weber, Mohamed F. Mokbel, Ravi Janardan, Scalable computational geometry in MapReduce. In *VLDB Journal*, 2019.
14. **Jie Xue**, Yuan Li, Ravi Janardan, On the expected diameter, width, and complexity of a stochastic convex-hull. In *Computational Geometry: Theory and Applications*, 2019.
15. Akash Agrawal, Yuan Li, **Jie Xue**, Ravi Janardan, The most-likely skyline problem for stochastic points. In *Computational Geometry: Theory and Applications* (CCCG'17 special issue), 2019.
16. **Jie Xue**, Yuan Li, Ravi Janardan, On the separability of stochastic geometric objects, with applications. In *Computational Geometry: Theory and Applications*, 2018.
17. Yuan Li, **Jie Xue**, Akash Agrawal, Ravi Janardan, On the arrangement of stochastic lines in  $\mathbb{R}^2$ . In *Journal of Discrete Algorithms*, 2017.

### Conference publications

1. Huairui Chu\*, Ajaykrishnan E S\*, Daniel Lokshantov\*, Anikait Mundhra\*, Thomas Schibler\*, Xiaoyang Xu\*, **Jie Xue\***, Parameterized approximation of rectangle stabbing. Accepted to the *34th Annual European Symposium on Algorithms (ESA)*, 2026.

2. Sujoy Bhore\*, Subhash Suri\*, **Jie Xue\***, Xiongxin Yang\*, Jiumu Zhu\*, Near-optimal dynamic data structures for maximum depth and Klee's measure of boxes. In the *53rd International Colloquium on Automata, Languages, and Programming (ICALP)*, 2026.
3. Sujoy Bhore\*, Chih-Hung Liu\*, Anurag Murty Naredla\*, Yakov Nekrich\*, Eunjin Oh\*, André van Renssen\*, Frank Staals\*, Haitao Wang\*, **Jie Xue\***, Visibility queries in simple polygons. In the *53rd International Colloquium on Automata, Languages, and Programming (ICALP)*, 2026.
4. Thomas Schibler\*, **Jie Xue\***, Jiumu Zhu\*, Approximating convex hulls via range queries. In the *42nd International Symposium on Computational Geometry (SoCG)*, 2026.
5. Daniel Lokshtanov\*, Fahad Panolan\*, Saket Saurabh\*, **Jie Xue\***, Meirav Zehavi\*, Fine-grained bounds for Courcelle's theorem. In the *58th Annual ACM Symposium on Theory of Computing (STOC)*, 2026.
6. Daniel Lokshtanov\*, Saket Saurabh\*, **Jie Xue\***, The sparsification lemma via measure and conquer. In the *9th SIAM Symposium on Simplicity in Algorithms (SOSA)*, 2026.
7. Thomas Schibler\*, Subhash Suri\*, **Jie Xue\***, Approximation and hardness of polychromatic TSP. In the *37th Canadian Conference on Computational Geometry (CCCG)*, 2025.
8. Sayan Bandyapadhyay\*, William Lochet\*, Daniel Lokshtanov\*, Dániel Marx\*, Pranabendu Misra\*, Daniel Neuen\*, Saket Saurabh\*, Prafullkumar Tale\*, **Jie Xue\***, Robust contraction decomposition for  $H$ -minor-free graphs and its applications. In the *52nd International Colloquium on Automata, Languages, and Programming (ICALP)*, 2025.
9. Subhash Suri\*, **Jie Xue\***, Xiongxin Yang\*, Jiumu Zhu\*, Dynamic maximum depth of geometric objects. In the *41st International Symposium on Computational Geometry (SoCG)*, 2025.
10. Shinwoo An\*, Eunjin Oh\*, **Jie Xue\***, Single-source shortest path problem in weighted disk graphs. In the *41st International Symposium on Computational Geometry (SoCG)*, 2025. Invited to SoCG special issue (regretfully declined).
11. Thomas Schibler\*, Subhash Suri\*, **Jie Xue\***, Embedding graphs as Euclidean  $k$ NN-graphs. In the *41st International Symposium on Computational Geometry (SoCG)*, 2025.
12. Daniel Lokshtanov\*, Fahad Panolan\*, Saket Saurabh\*, **Jie Xue\***, Meirav Zehavi\*, Subexponential parameterized algorithms for hitting subgraphs. In the *57th Annual ACM Symposium on Theory of Computing (STOC)*, 2025.
13. Daniel Lokshtanov\*, Fahad Panolan\*, Saket Saurabh\*, **Jie Xue\***, Meirav Zehavi\*, Efficiently finding and counting patterns with distance constraints in sparse graphs. In the *57th Annual ACM Symposium on Theory of Computing (STOC)*, 2025.
14. Shinwoo An\*, Eunjin Oh\*, **Jie Xue\***, Approximation algorithms for geometric multi-matching problem. In the *57th Annual ACM Symposium on Theory of Computing (STOC)*, 2025.
15. Daniel Lokshtanov\*, Fahad Panolan\*, Saket Saurabh\*, Roohani Sharma\*, **Jie Xue\***, Meirav Zehavi\*, Crossing number in slightly superexponential time. In the *36th ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2025.
16. Sayan Bandyapadhyay\*, Katie Clinch\*, William Lochet\*, Daniel Lokshtanov\*, Saket Saurabh\*, **Jie Xue\***, PTASes for Euclidean TSP with unit disk and unit square neighborhoods. In the *36th ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2025.
17. Daniel Lokshtanov\*, Abhishek Sahu\*, Saket Saurabh\*, Vaishali Surianarayanan\*, **Jie Xue\***, Parameterized approximation for capacitated  $d$ -hitting set with hard capacities. In the *36th ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2025.
18. Viktoriia Korchemna\*, Daniel Lokshtanov\*, Saket Saurabh\*, Vaishali Surianarayanan\*, **Jie Xue\***, Efficient approximation of hypertree width. In the *65th IEEE Symposium on Foundations of Computer Science (FOCS)*, 2024.
19. Daniel Lokshtanov\*, Fahad Panolan\*, Saket Saurabh\*, **Jie Xue\***, Meirav Zehavi\*, Bipartizing (pseudo-)disk graphs: approximation with a ratio better than 3. In the *27th Interna-*

- tional Conference on Approximation Algorithms for Combinatorial Optimization Problems (APPROX)*, 2024.
20. Shinwoo An\*, Eunjin Oh\*, **Jie Xue\***, Sparse outerstring graphs have logarithmic treewidth. In the *32nd Annual European Symposium on Algorithms (ESA)*, 2024.
  21. Sayan Bandyapadhyay\*, **Jie Xue\***, An  $O(n \log n)$ -time approximation scheme for Euclidean many-to-many matching. In the *40th International Symposium on Computational Geometry (SoCG)*, 2024. Winner of the Best Paper Award.
  22. Haitao Wang\*, **Jie Xue\***, Algorithms for halfplane coverage and related problems. In the *40th International Symposium on Computational Geometry (SoCG)*, 2024.
  23. Kyungjin Cho\*, Eunjin Oh\*, Haitao Wang\*, **Jie Xue\***, Optimal algorithm for the planar two-center problem. In the *40th International Symposium on Computational Geometry (SoCG)*, 2024. Invited to TheoretiCS.
  24. Daniel Lokshantov\*, Fahad Panolan\*, Saket Saurabh\*, **Jie Xue\***, Meirav Zehavi\*, A 1.9999-approximation algorithm for vertex cover in string graphs. In the *40th International Symposium on Computational Geometry (SoCG)*, 2024.
  25. Timothy M. Chan\*, Qizheng He\*, **Jie Xue\***, Enclosing points with geometric objects. In the *40th International Symposium on Computational Geometry (SoCG)*, 2024.
  26. Sayan Bandyapadhyay\*, William Lochet\*, Daniel Lokshantov\*, Saket Saurabh\*, **Jie Xue\***, Euclidean bottleneck Steiner tree is fixed-parameter tractable. In the *35th ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2024.
  27. Chinmay Sonar\*, Subhash Suri\*, **Jie Xue\***, Fault tolerance in Euclidean committee selection. In the *31st Annual European Symposium on Algorithms (ESA)*, 2023.
  28. Sayan Bandyapadhyay\*, William Lochet\*, Saket Saurabh\*, **Jie Xue\***, Minimum-membership geometric set cover, revisited. In the *39th International Symposium on Computational Geometry (SoCG)*, 2023.
  29. Akanksha Agrawal\*, Tanmay Inamdar\*, Saket Saurabh\*, **Jie Xue\***, Clustering what matters: optimal approximation for clustering with outliers. In the *37th AAAI conference on Artificial Intelligence (AAAI)*, 2023. Selected as AAAI Distinguished Paper.
  30. Daniel Lokshantov\*, Fahad Panolan\*, Saket Saurabh\*, **Jie Xue\***, Meirav Zehavi\*, A framework for approximation schemes on disk graphs. In the *34th ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2023.
  31. Rong Gu, Han Li, Haipeng Dai, Wenjie Huang, **Jie Xue**, Meng Li, Jiaqi Zheng, Haoran Cai, Yihua Huang, Guihai Chen, ShadowAQP: efficient approximate group-by and join query via attribute-oriented sample size allocation and data generation. In the *49th International Conference on Very Large Data Bases (VLDB)*, 2023.
  32. Chinmay Sonar\*, Subhash Suri\*, **Jie Xue\***, Multiwinner elections under minimax Chamberlin-Courant rule in Euclidean space. In the *31st International Joint Conference on Artificial Intelligence (IJCAI)*, 2022.
  33. Sayan Bandyapadhyay\*, William Lochet\*, Daniel Lokshantov\*, Saket Saurabh\*, **Jie Xue\***, True contraction decomposition and almost ETH-tight bipartization for unit-disk graphs. In the *38th International Symposium on Computational Geometry (SoCG)*, 2022.
  34. Neeraj Kumar\*, Daniel Lokshantov\*, Saket Saurabh\*, Subhash Suri\*, **Jie Xue\***, Point separation and obstacle removal by finding and hitting odd cycles. In the *38th International Symposium on Computational Geometry (SoCG)*, 2022. Invited to SoCG special issue (regretfully declined).
  35. Sayan Bandyapadhyay\*, William Lochet\*, Daniel Lokshantov\*, Saket Saurabh\*, **Jie Xue\***, Subexponential parameterized algorithms for cut and cycle hitting problems on H-minor-free graphs. In the *33rd ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2022.
  36. Timothy M. Chan\*, Qizheng He\*, Subhash Suri\*, **Jie Xue\***, Dynamic geometric set cover, revisited. In the *33rd ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2022.

37. Daniel Lokshantov\*, Fahad Panolan\*, Saket Saurabh\*, **Jie Xue\***, Meirav Zehavi\*, Subexponential parameterized algorithms on disk graphs. In the *33rd ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2022.
38. Daniel Lokshantov\*, Saket Saurabh\*, Subhash Suri\*, **Jie Xue\***, An ETH-tight algorithm for multi-team formation. In the *41st Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS)*, 2021.
39. Daniel Lokshantov\*, Subhash Suri\*, **Jie Xue\***, Efficient algorithms for least square piecewise polynomial regression. In the *29th Annual European Symposium on Algorithms (ESA)*, 2021.
40. Zhenyu Pan, **Jie Xue**, Tingjian Ge, Intuitive searching: an approach to search the decision policy of a Blackjack agent. In the *6th International Congress on Information and Communication Technology (ICICT)*, 2021.
41. Daniel Lokshantov\*, Chinmay Sonar\*, Subhash Suri\*, **Jie Xue\***, Fair covering of points by balls. In the *32nd Canadian Conference on Computational Geometry (CCCG)*, 2020.
42. Pankaj K. Agarwal\*, Hsien-Chih Chang\*, Subhash Suri\*, Allen Xiao\*, **Jie Xue\***, Dynamic geometric set cover and hitting set. In the *36th International Symposium on Computational Geometry (SoCG)*, 2020. Invited to SoCG special issue (regretfully declined).
43. Haitao Wang\*, **Jie Xue\***, Improved algorithms for the bichromatic 2-center problem for pairs of points. In the *16th Algorithms and Data Structures Symposium (WADS)*, 2019.
44. Timothy Chan\*, Rahul Saladi\*, **Jie Xue\***, Range closest-pair search in higher dimensions. In the *16th Algorithms and Data Structures Symposium (WADS)*, 2019. Invited to WADS special issue.
45. **Jie Xue**, Yuan Li, Rahul Saladi, Ravi Janardan, Searching for the closest-pair in a query translate. In the *35th International Symposium on Computational Geometry (SoCG)*, 2019. Invited to SoCG special issue.
46. Haitao Wang\*, **Jie Xue\***, Near-optimal algorithms for shortest paths in weighted unit disk graphs. In the *35th International Symposium on Computational Geometry (SoCG)*, 2019. Invited to SoCG special issue.
47. **Jie Xue**, Colored range closest-pair problem under general distance functions. In the *30th ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2019.
48. **Jie Xue**, Yuan Li, Ravi Janardan, Approximate range closest-pair queries. In the *30th Canadian Conference on Computational Geometry (CCCG)*, 2018. Invited to CCCG special issue.
49. **Jie Xue**, Yuan Li, Rahul Saladi, Ravi Janardan, New bounds for range closest-pair problems. In the *34th International Symposium on Computational Geometry (SoCG)*, 2018.
50. Zhenyu Pan, **Jie Xue**, Yang Gao, Honghao Wang, Guanling Chen, Revealing the relations between learning behaviors and examination scores via a prediction system. In the *2nd International Conference on Computer Science and Artificial Intelligence (CSAI)*, 2018.
51. **Jie Xue**, Yuan Li, Ravi Janardan, On the expected diameter, width, and complexity of a stochastic convex-hull. In the *15th Algorithms and Data Structures Symposium (WADS)*, 2017.
52. **Jie Xue**, Yuan Li, Stochastic closest-pair problem and most-likely nearest-neighbor search in tree spaces. In the *15th Algorithms and Data Structures Symposium (WADS)*, 2017.
53. Akash Agrawal, Yuan Li, **Jie Xue**, Ravi Janardan, The most-likely skyline problem for stochastic points. In the *29th Canadian Conference on Computational Geometry (CCCG)*, 2017. Invited to CCCG special issue.
54. **Jie Xue**, Yuan Li, Ravi Janardan, On the separability of stochastic geometric objects, with applications. In the *32nd International Symposium on Computational Geometry (SoCG)*, 2016.

## Manuscripts

1. Akanksha Agrawal\*, Tanmay Inamdar\*, **Jie Xue\***, Algorithms for simultaneous flip distance between triangulations of polygons. Submitted.
2. Sayan Bandyapadhyay\*, William Lochet\*, Saket Saurabh\*, **Jie Xue\***, FPTAS for polygonal multi-robot motion planning. Submitted.
3. Madhumita Kundu\*, Daniel Lokshtanov\*, Saket Saurabh\*, **Jie Xue\***, Capacitated set cover, hitting set, and domination in the semiladder-free setting. Submitted.
4. Zdeněk Dvořák\*, Daniel Lokshtanov\*, Fahad Panolan\*, Saket Saurabh\*, **Jie Xue\***, Meirav Zehavi\*, Efficient approximation for subgraph-hitting problems in sparse graphs and geometric intersection graphs. Submitted.
5. Sayan Bandyapadhyay\*, William Lochet\*, Daniel Lokshtanov\*, Saket Saurabh\*, Chinmay Sonar\*, Vaishali Surianarayanan\*, **Jie Xue\***, How to fix a broken metric: a linear kernel, tight bounds, and tree metrics. Submitted.
6. Daniel Lokshtanov\*, Fahad Panolan\*, Saket Saurabh\*, **Jie Xue\***, Meirav Zehavi\*, Fine-grained MSO model checking for cliquewidth. Submitted.
7. Yuxi Liu\*, Saket Saurabh\*, Kangyi Tian\*, Mingyu Xiao\*, **Jie Xue\***, Constant-factor FPT approximation for rectangle hitting set and rectangle set cover. Submitted.
8. Sayan Bandyapadhyay\*, William Lochet\*, Saket Saurabh\*, **Jie Xue\***, Euclidean capacitated sum-of-radii clustering: hardness and approximation. Submitted.
9. Huairui Chu\*, Ajaykrishnan E S\*, Daniel Lokshtanov\*, Viktoriia Korchemna\*, Saket Saurabh\*, Thomas Schibler\*, Xiaoyang Xu\*, **Jie Xue\***, Xiongxin Yang\*, Exponential time approximation schemes for layout problems. Submitted.
10. Ajaykrishnan E S\*, Daniel Lokshtanov\*, Saket Saurabh\*, **Jie Xue\***, Algorithms for min-color path with bounded color-degree. In progress.
11. Sujoy Bhore\*, **Jie Xue\***, Xiongxin Yang\*, Jiumu Zhu\*, Simple and efficient algorithms for approximating maximum depth of unit disks and unit balls. In progress.
12. Sujoy Bhore\*, **Jie Xue\***, Xiongxin Yang\*, Jiumu Zhu\*, Approximate diameter in dynamic unit disk graphs. In progress.
13. **Jie Xue**, Yuan Li, On dominance-free samples of a (colored) stochastic dataset. In progress.

## TALKS

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### Invited talks

- Talk at Chengdu A&L Seminar (June 2026):  
Fine-grained bounds for Courcelle's theorem.
- Talk at NYU Shanghai (April 2026):  
Efficient algorithms for geometric and graph problems.
- Tutorial talk at MTPC Workshop (April 2026):  
Parameterized problems and algorithms in computational geometry.
- Talk at MTPC Workshop 2026 (April 2026):  
Fine-grained bounds for Courcelle's theorem.
- Talk at UCSB (Jan. 2026):  
Dynamic maximum depth and Klee's measure.
- Talk at NYU Theory Seminar (May 2025):  
Subexponential parameterized algorithms for hitting subgraph.
- Talk at Courant Geometry Seminar (April 2025):  
Efficient approximation algorithms for geometric many-to-many matching.
- Talk at Shanghai Theory Day (March 2025):  
Subexponential parameterized algorithms for hitting subgraph.

- Talk at Lorentz Workshop (Feb. 2025):  
Efficient approximation algorithms for geometric many-to-many matching.
- Talk at Dagstuhl 24411 (Oct. 2024):  
New developments in parameterized computational geometry.
- Talk at IJTCS-FAW 2024:  
A 1.9999-approximation algorithm for vertex cover in string graphs.
- Talk at CCF National Conference of TCS 2024:  
An  $O(n \log n)$ -time approximation scheme for Euclidean many-to-many matching.
- Talk at UCSB (Oct. 2023):  
Vertex deletion on disk graphs.
- Talk at Nanjing University (Aug. 2022):  
Vertex deletion on disk graphs.
- Talk at NYU Tandon (Online, Nov. 2021):  
Efficient algorithms and data structures for geometric computing.
- Talk at NYU Shanghai (Online, March 2021):  
Efficient algorithms and data structures for geometric computing.
- Talk at Nanjing University (Online, Dec. 2020):  
Efficient algorithms and data structures for geometric computing.
- Talk at Nanjing University (Aug. 2018):  
Range closest-pair search.

#### Conference presentations

- Presentation at SoCG 2024 (Online):
  1. An  $O(n \log n)$ -time approximation scheme for Euclidean many-to-many matching.
  2. A 1.9999-approximation algorithm for vertex cover in string graphs.
- Presentation at SoCG 2023:  
Minimum-membership geometric set cover, revisited.
- Presentation at SODA 2023 (Online):  
A framework for approximation schemes on disk graphs.
- Presentation at SoCG 2022 (Online):  
Point separation and obstacle removal by finding and hitting odd cycles.
- Presentation at SODA 2022 (Online):
  1. Subexponential parameterized algorithms for cut and cycle hitting problems on H-minor-free graphs.
  2. Subexponential parameterized algorithms on disk graphs.
- Presentation at ESA 2021 (Online):  
Efficient algorithms for least square piecewise polynomial regression.
- Presentation at SoCG 2020:  
Dynamic geometric set cover and hitting set.
- Presentation at SoCG 2019:
  1. Searching for the closest-pair in a query translate.
  2. Near-optimal algorithms for shortest paths in weighted unit disk graphs.
- Presentation at SODA 2019 and FWCG 2018:  
Colored range closest-pair problem under general distance functions.
- Presentation at CCCG 2018:  
Approximate closest-pair search.
- Presentation at SoCG 2018:  
New bounds for range closest-pair problems.
- Presentation at Young Researcher Forum in CG Week 2018:  
Searching for the closest-pair in a convex polygonal translate.

- Presentation at WADS 2017:
  1. Stochastic closest-pair problem and most-likely NN search in tree spaces.
  2. On the expected diameter, width, and complexity of a stochastic convex-hull.
- Presentation at SoCG 2016:  
On the separability of stochastic geometric objects, with applications.

## STUDENTS

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### PhD students

- Jiumu Zhu (2024-now)

### Undergraduate students advised

- Xiongxin Yang (2024-2025)
- Junqiu Hu (2025-present)

## PROFESSIONAL ACTIVITIES AND SERVICES

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### Workshop organizer

- Parameterized Algorithms for Geometric Problems (at SoCG 2023)

### Editorial board of journals

- Journal of Computational Geometry (2026-Current)
- DCG special issue of SoCG'25 (2025)

### Program committee of conferences

- ICALP 2027
- SOSA 2027
- CCCG 2026
- SoCG 2025
- ESA (Track A) 2025
- WADS 2023, 2025
- IPEC 2025
- STACS 2024, 2027
- FAW 2022
- CG Week YRF 2021

### Reviewer for journals

- Journal of the ACM
- SIAM Journal on Computing
- Discrete & Computational Geometry
- Journal of Computational Geometry
- Algorithmica
- Computational Geometry: Theory and Applications
- Theoretical Computer Science
- Computing in Geometry and Topology
- Journal of Combinatorial Optimization
- International Journal of Computational Geometry and Applications
- VLDB Journal

### **Reviewer for conferences**

- STOC 2023, 2026
- FOCS 2021, 2024, 2025, 2026
- SODA 2021, 2022, 2023, 2024, 2025, 2026, 2027
- SoCG 2017, 2020, 2022, 2023, 2024, 2026
- ICALP 2021, 2022, 2024, 2025, 2026
- ESA 2022, 2023, 2026
- APPROX 2023, 2026
- ISAAC 2018, 2020, 2021, 2023, 2024, 2026
- WADS 2017
- SWAT 2020
- FSTTCS 2021, 2024
- MFCS 2017
- IPEC 2023, 2026
- FAW 2019
- COCOON 2023
- COCOA 2021

### **Services at NYU Shanghai**

- CSDSE faculty search committee 2025-2026
- PC for Capstone projects 2021, 2022, 2023
- Mentor for DURF projects 2022, 2023, 2026

## **AWARDS**

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- SoCG Best Paper Award (2024)
- AAAI Distinguished Paper Award (2023)
- Shanghai Leading Talents Program (2023)
- SoCG Travel Award (2016, 2018, 2019)
- SIAM Travel Award (2018)
- Doctoral Dissertation Fellowship at University of Minnesota (2018-2019)

## **TEACHING**

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### **At NYU Shanghai:**

- CSCI-SHU 2314 - Discrete Math (Fall 2024, Fall 2025, Spring 2026, Fall 2026)
- CSCI-SHU 220 - Algorithms (Spring 2022, Spring 2023, Spring 2024)
- CSCI-SHU 210 - Data Structures (Spring 2023)
- CSCI-SHU 11 - Introduction to Computer Programming (Fall 2021, Fall 2022)

### **At University of Minnesota (as Teaching Assistant):**

- CSci 5421 - Advanced Algorithms and Data Structures (Spring 2015, Fall 2017, Spring 2018)
- CSci 4011 - Formal Languages and Automata Theory (Spring 2016, Fall 2016, Spring 2017)
- CSci 2011 - Discrete Structures of Computer Science (Fall 2015)

## LANGUAGES

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- Chinese (traditional preferred)
- English (US)

## HOBBIES

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- Calligraphy
- Poetry (traditional)
- Reading
- Table-tennis
- Video games (RPG, RTS, MOBA, etc.)